



22CST61 - COMPILER DESIGN							
Programme & Branch	B.E. - Computer Science and Engineering	Sem.	Category	L	T	P	Credit
Prerequisites	NIL	6	PC	3	0	0	3
Preamble	This course provides an insight into the compiler construction process as well as the design techniques for the given programming language.						
Unit – I	Lexical Analysis						9
Introduction – Language Processors – The structure of a compiler – Lexical Analysis – The Role of the Lexical Analyzer – Input Buffering – Specification of Tokens – Recognition of Tokens – The Lexical-Analyzer Generator – Lex – Finite Automata – From Regular Expressions to Automata.							
Unit – II	Syntax Analysis						9
Introduction – Context-Free Grammars – Writing a Grammar – Top-Down Parsing – Bottom-Up parsing – Operator Precedence Parser – Introduction to LR Parsing: Simple LR – More Powerful LR Parsers – Parser Generators.							
Unit – III	Syntax - Directed Translation and Intermediate Code Generation						9
Syntax-Directed Translation – Evaluation orders for SDDs – Intermediate Code Generation – Variants of syntax trees – Three Address Code – Types and Declarations – Translation of Expressions – Control Flow – Backpatching – Switch Statements – Procedure calls.							
Unit – IV	Machine Independent Optimizations						9
Basic Blocks and Flow Graphs – Optimization of Basic Blocks– Peephole Optimization – The Principal Sources of Optimization – Introduction to Data-Flow Analysis – loops and flow graphs.							
Unit – V	Code Generation and Storage Management						9
Issues in the design of a code generation – The target Language – Addresses in the Target code – A simple code Generator – Run-Time Environments: Storage organization – Stack allocation of space – Heap Management – Introduction to garbage collection.							
							Total:45
TEXT BOOK:							
1.	Alfred V. Aho, Monica S. Lam, Ravi Sethi and Jeffrey D. Ullman, “Compilers: Principles, Techniques and Tools”, 2nd Edition, Pearson India Education Pvt. Ltd., 2014.						
REFERENCES:							
1.	Srikant Y.N. and Priti Shankar, “The Compiler Design Handbook: Optimizations and Machine Code Generation”, 2nd Edition, CRC Press, 2008						
2.	Kenneth C. Louden, “Compiler Construction – Principles and Practice”, 1st edition, PWS Publishing. 1997						



COURSE OUTCOMES: On completion of the course, the students will be able to												BT Mapped (Highest Level)			
CO1	make use of regular expression to perform lexical analysis of the source program												Applying (K3)		
CO2	design a syntax-analysis tool for the given grammar												Applying (K3)		
CO3	develop intermediate code for the source program												Applying (K3)		
CO4	employ optimization techniques for the given intermediate code												Applying (K3)		
CO5	apply suitable storage allocation technique to generate the target code												Applying (K3)		
Mapping of COs with POs and PSOs															
COs/ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	
CO1	3	2	1		1										
CO2	3	2	1		1										
CO3	3	2	1												
CO4	3	2	1												
CO5	3	2	1												
1 – Slight, 2 – Moderate, 3 – Substantial, BT- Bloom's Taxonomy															
ASSESSMENT PATTERN – THEORY															
Test / Bloom's Category*		Remembering (K1) %		Understanding (K2) %		Applying (K3) %		Analyzing (K4) %		Evaluating (K5) %		Creating (K6) %		Total %	
CAT1		-		35		65								100	
CAT2		-		25		75								100	
CAT3		-		60		40								100	
ESE		-		35		65								100	
* ±3% may be varied (CAT 1, 2, 3 – 50 marks & ESE – 100 marks)															